

VISUALIZING DIFFICULTY IN SIGHT-SINGING PRACTICE WITH COLOR-CODED SCORES

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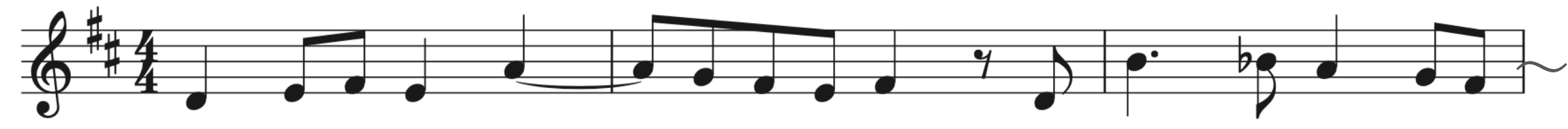
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Introduction

• **Sight-singing**: the ability to sing a piece of written music accurately after a short preview time

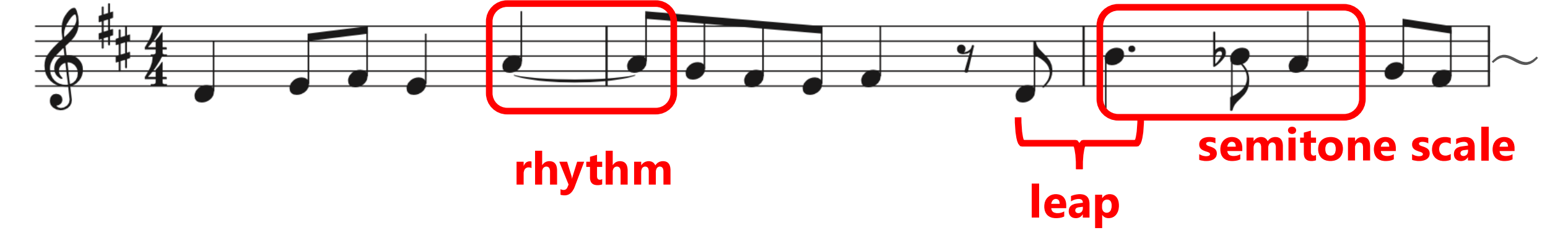
• Sight-singing strategy example: **Scanning music to find the challenging spots** [1]



PREVIEW the score

SING without practice

- A fundamental skill for all musicians
- Often difficult for learners to acquire (a **common educational challenge**)



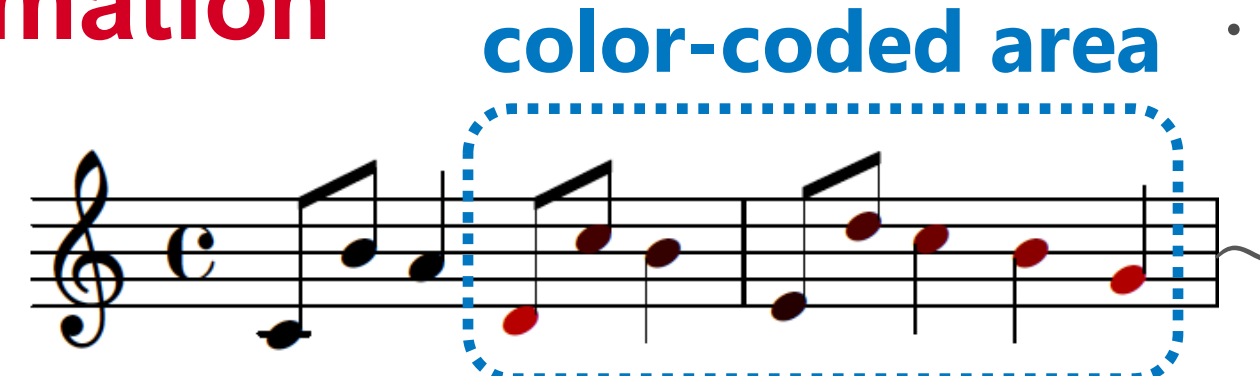
However, some learners sight-sing without using any specific strategies [2]. For these learners, visually marking difficult spots on the score could help them engage more strategically during the preview time.

Color-Coded Score

Color-Coded Score with Difficulty Information

• Color-Code difficulty visualization: **black-to-red** gradient

easy difficult



• **Objective difficulty based on statistical data**

• Cost-effective scalable evaluation and application into online education

• SPED: A large-scale dataset of sight-singing performance evaluations [3]

Difficulty Estimation Model

• Definition of note difficulty: the error rate across multiple performances

$$\text{diff}_{n,k} = 1 - \frac{\sum_{i=1}^{N_n} (p_{n,k,i} \cdot r_{n,k,i})}{N_n}$$

N_n : the number of performances of task n
 $p_{n,k,i}, r_{n,k,i}$: pitch and rhythm correctness (binary score) of the k -th note in the i -th performance

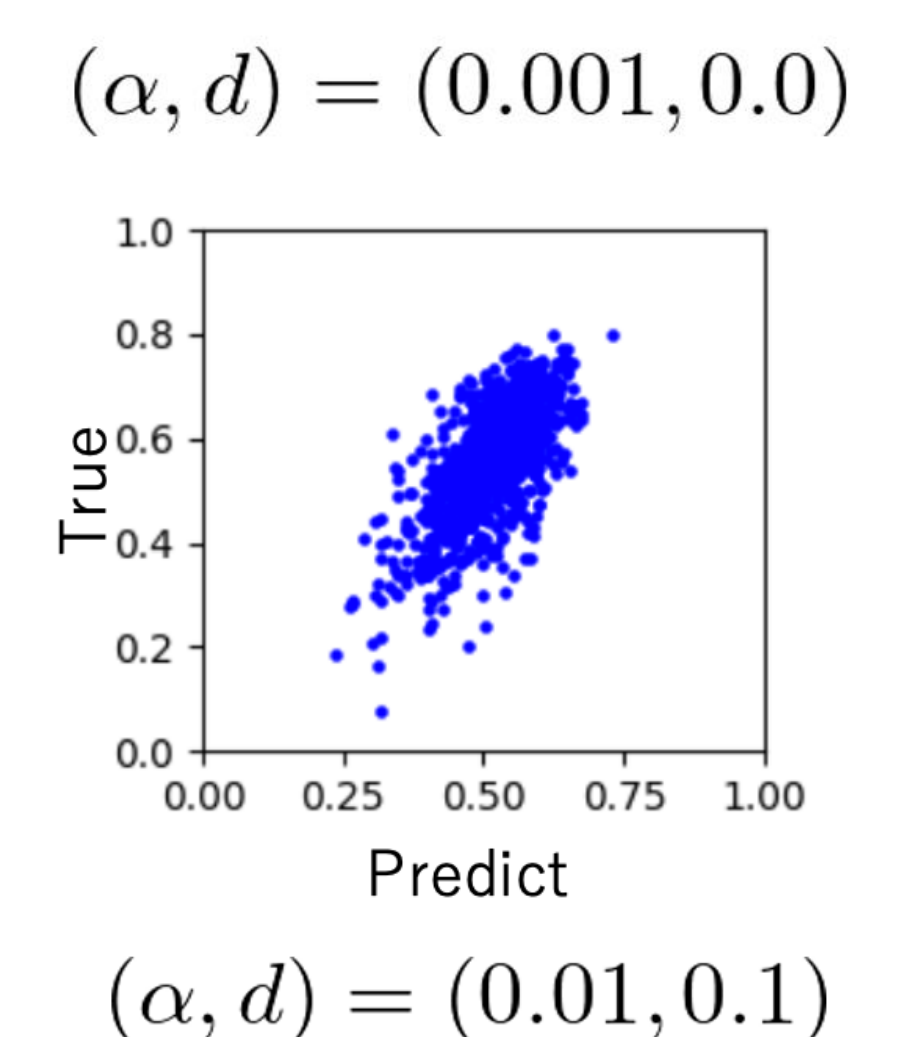
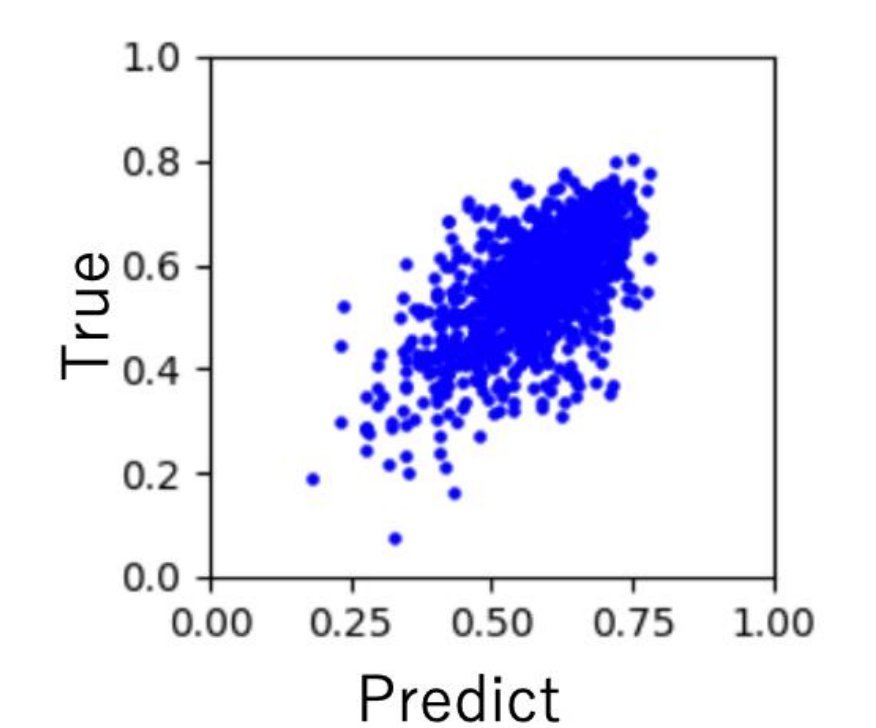
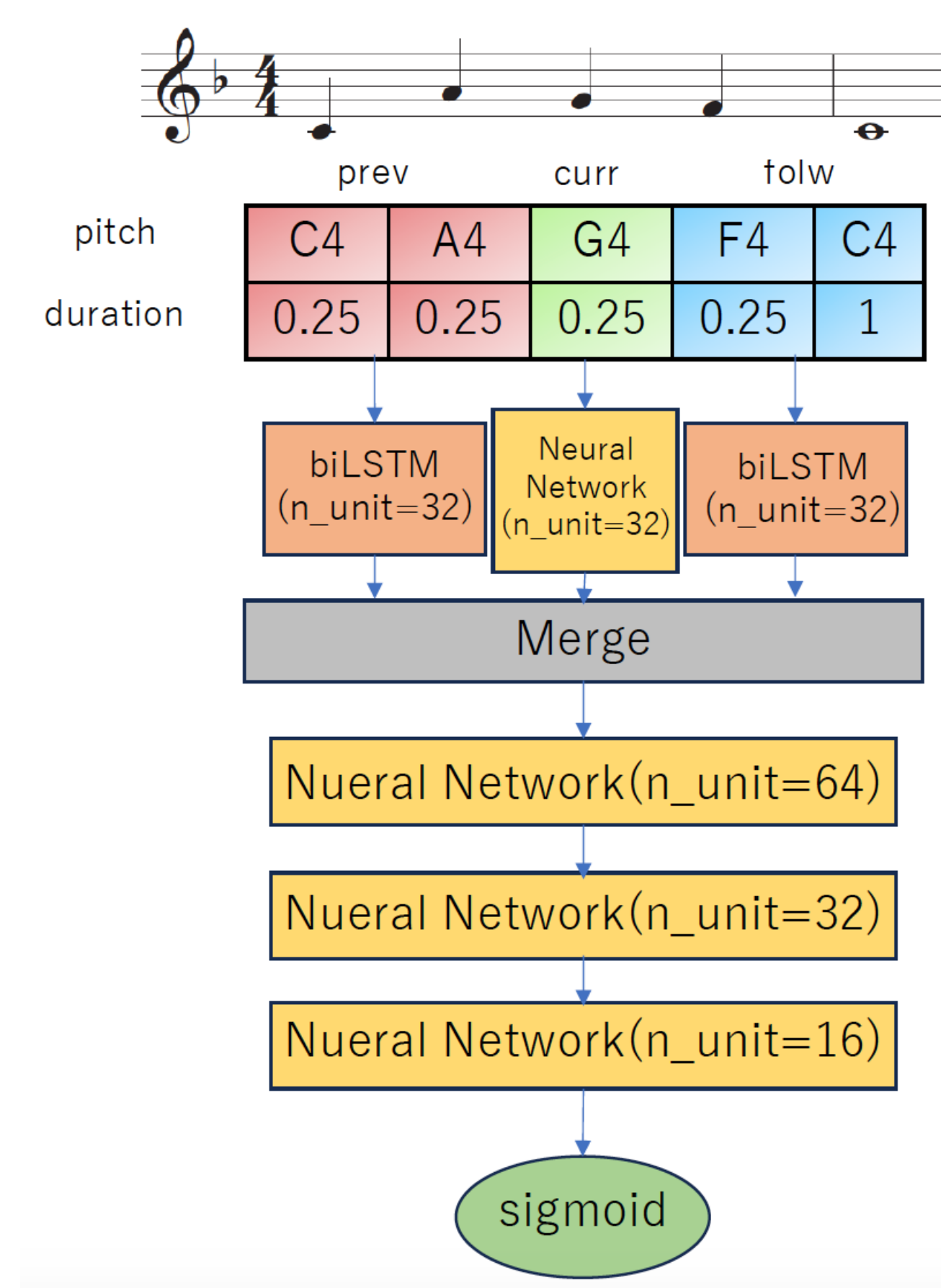
• Model architecture: bi-directional LSTM model (cf. DeepBach [4])

• Training conditions

- Input: sequences of three preceding and succeeding notes
- Using only tasks with $N_n \geq 500$ for statistical reliability
- 100 epochs
- Learning rate $\alpha = 0.1, 0.01, 0.001$; dropout $d = 0.0, 0.1, 0.2, 0.3$

• Test results

- **Best correlation coefficient $r = 0.689$** ($(\alpha, d) = (0.01, 0.1)$)
- Worst correlation coefficient $r = 0.616$ ($(\alpha, d) = (0.001, 0.0)$)
- Scatter plots comparing predicted and ground-truth scores



Generating Color-Coded Scores

• Samples of color-coded scores with the sight-singing tasks from a Japanese textbook [5]

- Normalization of difficulty values: using min-max scaling within each task
- Treatment of rests as extensions of the preceding note

【Task 1】

(Good) B $\flat$ (m. 2) : difficult descent from C to B $\flat$ following B natural

(Limitation) B (m. 2) may be more challenging (non-diatonic note in F major).

(Limitation) E-E $\flat$ -D-E (m. 7) also appears to be more difficult.

【Task 2】

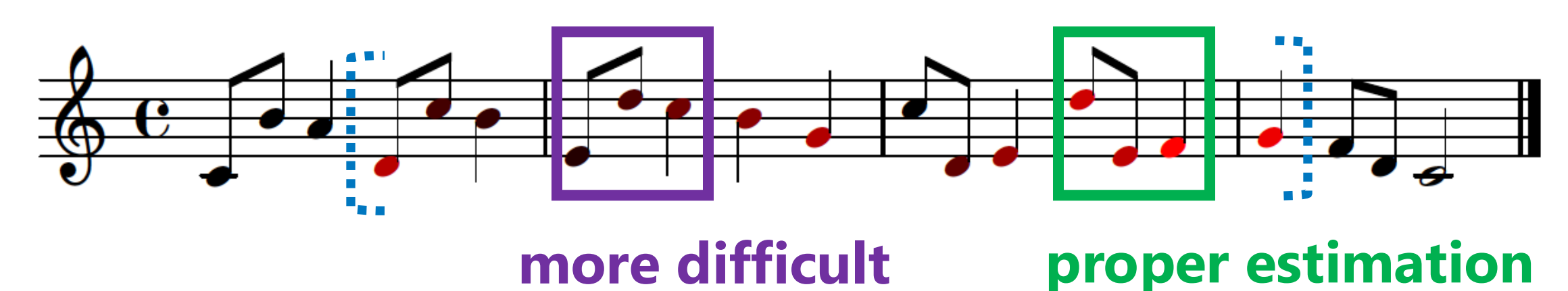
(Good) D-E-F (m. 3) : the difficulty of minor-seventh leap

(Limitation) There are other similarly challenging leaps.

Task 1



Task 2



Conclusion

This study explored new **color-coded scores** based on difficulty estimated using a bi-directional LSTM model. The model showed moderate accuracy.

• **Future Work**

- Conduct user studies with novice learners to assess the pedagogical validity and its usefulness in sight-singing practice.
- Extend the model to Identify not only how difficult a note is but why.

References

- [1] G. Fournier, M. T. M. Sala, F. Dubé, and S. O'Neill, "Cognitive strategies in sight-singing: The development of an inventory for aural skills pedagogy," *Psychology of Music*, vol. 47, no. 2, pp. 270–283, 2019.
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- [5] T. Akaishi, H. Suzuki, R. Nakajima, K. Nakano, K. Yamamoto, and I. Kasuya, *Graded Sight Singing Studies*. Tokyo, Japan: Zen-On Music Company Limited, 2008.